

Mohammad Sefid | Curriculum Vitae

Department of Electrical Eng. K.N.Toosi University of Technology, Tehran, Iran
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- Master's level research in RIS-aided Integrated Sensing and Communication (ISAC) systems with Cooperative NOMA through Energy Harvesting from SWIPT operation and Optimize with Deep Reinforcement Learning (DRL), Alternating Optimization (AO) and Metaheuristic techniques

Education

- **K. N. Toosi University of Technology** **Tehran, Iran**
MSc Electrical, Electronics and Communications Engineering 2023–2026

Notable Courses and Projects:
 - Performance improvement of RIS-assisted ISAC systems
 - Machine Learning ([Code](#))
 - Numerical Optimization ([Code](#))
 - Digital Signal Processing ([Code](#))
 - Adaptive Filters ([Code](#))
- **University of Applied Science and Technology** **Iran**
Bachelor of Science, Electrical and Electronics Engineering

Notable Projects:
 - A Review of Photovoltaic (PV) Technologies: System Components, Architectures, and Future Research Directions
- **Arak University of Technology** **Iran**
Associate Degree, Electrical and Electronics Engineering

Publications

- IRS-Assisted Cooperative NOMA with SWIPT in ISAC Systems: A Comparison of Half-Duplex and Full-Duplex Modes [In progress]

Professional Experience

- **K. N. Toosi University of Technology** **Tehran, Iran**
Teaching Assistant Feb 2025- Jan 2026
 - Numerical Optimization
 - Stochastic Processes
 - Estimation Theory
 - Information Theory
- **Zamyad Co.** **Tehran, Iran**
Intern Jul-Sep 2018
 - Developed knowledge of automotive electrical and electronic systems
 - Provided electrical and electronic support for quality control laboratories and manufacturing lines
 - Evaluated safety measures and contributed to the installation of grounding (earthing) systems for technical lines
- **Manasazan Sadid** **Tehran, Iran**
Intern Apr-June 2021

- Gained practical experience in the planning and design of renewable energy power plants
- Supported the installation of energy systems, including off-grid solar power plants
- Developed understanding of installation project management and logistics coordination

○ **Ariosunshade**
Technical content creator

Tehran, Iran
Sep 2017-Mar 2018

- Translated technical energy-related articles from English into Persian
- Conducted literature review to identify state-of-the-art advancements in the energy sector

Technical Tools and Methods

- **Programming:** Python | MATLAB | GitHub
- **Data Analysis:** Optimization | Machine Learning
- **Writing Tools:** L^AT_EX | Microsoft Office

Conferences

- Conference on Electrical and Computer Innovations, K. N. Toosi University of Technology, Feb 2025
 Presented my research on "Applications of STAR RIS in Communication Systems"

Research Interests

- Joint Optimization of Integrated Communication and Sensing (ISAC) Performance in Next-Generation Wireless Networks
- Simultaneous Wireless Information and Power Transfer (SWIPT) for Cooperative Non-Orthogonal Multiple Access (C-NOMA) Systems
- Optimization of Reconfigurable Intelligent Surfaces (RIS) to Enhance Beamforming of Wireless Signals
- Deep Reinforcement Learning (DRL) optimization for adaptive wireless Systems

Extra-Curricular Activities and Hobbies

- Half Marathon Runner
- Recreational Cyclist
- Regular Workout